Exercise Three

1. A discrete random variable X has the following probability distribution.

X	-2	-1	0	1	2
P(X)	K	0.2	2K	2K	0.1

   a) Find k
   b) Find the expected value of the random variable X.

2. A random variable follows the probability distribution given below;

X	0	1	2	3	4
P(X)	0.12	0.23	K	0.2	0.1

   a) Obtain the value of k,

   b) Compute the expected value of X

   c) Find $Var(X)$

3. A random variable X has the following pdf

$$f(x) = \begin{cases} cx^2 & 0 < x < 3 \\ 0 & otherwise \end{cases}$$

   a) Find the constant c
   b) Compute $P(1 < x < 2)$
   c) Find $E(X)$

4. In a lottery there are 200 prizes of Sh5, 20 prizes of Sh 25, and 5 prizes of Sh 100. Assuming that 10,000 tickets are to be issued and sold, what is the expected fair price to pay for a ticket?
5. A continuous random variable X has probability density given by

$$f(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & otherwise \end{cases}$$

   a) Find $E(X)$
   b) Find $Var(X)$

6. A game is played as follows; throw a fair four sided dice and score eight times the number that faces down unless it is a four, you are given a second chance in which you score only four times that faces down. Let X be a random variable denoting the score for each player. Represent this information on a tree diagram showing the value of X and the corresponding probability, hence or otherwise find the mean of the scores
7. The random variable X has the following p.d.f:

$$f(x)=\begin{cases}\frac{1}{7}\left(4-3x^2\right), & -1\le x\le 1\\ \frac{1}{7}, & 1<x\le 2\\ 0, & \text{otherwise}\end{cases}.$$

i. Sketch the p.d.f. (2 marks)

ii. Find the probability $P(0.5\le x\le 1.5)$. (4 marks)

iii. Find $E(X)$. (2 marks)

8. The probability function of some discrete variables is given by

$$f(x)=\begin{cases}k(x-1), & x=2,3\\ k(8-x), & x=4,5,6\\ 0, & \text{otherwise}\end{cases}$$

Find:

a) The value of k.

b) $P(x\le 4)$.

c) $E(X)$

d) $Var(X)$

9. A discrete random variable X has the probability function

$$P(X=x)=\begin{cases}k(1-x)^2 & x=-1,0,1\ and\ 2\\ 0 & otherwise\end{cases}$$

Find

i) k

ii) $E(X)$ and $Var(X)$

iii) $Var(1-3X)$